A PRISMA-Guided Systematic Review of Machine Learning Algorithms Using Physiological Signals for Mental Health Screening in Learning Institutions

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***Abstract*—Reliable mental health screening in educational settings remains a challenge. This PRISMA 2020-guided systematic review examined physiological signals,machine learning algorithms, mental health outcomes and reported model performances in student focused studies. PubMed and IEEE Xplore were the two main study sources searched for studies published from 2018 to 2026. Older studies were included when they provided vital information on signals such as pupil diameter. In total, 17 studies met the eligibility criteria and were included in the review.**

The studies applied algorithms such as Random Forest, Decision Trees and Logistic Regression, examined signals such as heart rate and pupil diameter and focused on mental health outcomes like stress, anxiety and depression. Accuracy was the most reported metric across the studies though metrics like F1 score and Area Under the Curve (AUC-ROC) were reported. Diversity in datasets, signals, models and metrics also limited the evidence.

***Index Terms*—Physiological Signals, Machine Learning, PRISMA, Stress, Anxiety, Depression, Random Forest, Explainable AI,**

I. Introduction

Mental Health disorders are a growing global health concern. 700,000 suicide cases occur globally every year and 77% of these happen in low and middle income countries like Uganda [1].

Within Uganda mental health is a huge concern among the young people. Studies report that 23.6% of Ugandan children live with depression [2]. Most are school going yet physical conditions like height, weight BMI are measured but rarely is mental health assessed. As a result stress, anxiety and depression remain undetected increasing the risk of poor mental health outcomes.

Despite this, mental health screening in schools is low. Counselors exist but underutilized. Traditional methods like questionnaires have been used though are prone to bias as students maybe reluctant to disclose the truth due to fear. Advanced methods such as electroencephalography(EEG) are reliable, but expensive and need professional analysis. Other studies proposed using wearable sensors but were prone to noise by movement which could disrupt data collection [3]. These limitations make reliable screening difficult.

Recent studies have shown pupil dilation and constriction as possible biomarkers for brain activity. Pupil dynamics are controlled by the autonomic nervous system,which has two branches namely sympathetic and parasympathetic systems. The sympathetic is triggered during stress and pupils dilate while the parasympathetic induces constriction as body relaxes [4].The hypothalamus connects the endocrine system and nervous system. Changes in emotional state lead to cortisol release a stress hormone raising heart rate [5]. These signals may therefore be useful for screening Although physiological signals and machine learning models have been explored for mental health screening, the evidence remains dispersed across different signals , outcomes and evaluation methods. This systematic review aims to bridge that gap by identifying the different machine learning models and outcomes reported, and further synthesizing their performance metrics and methodological limitations. It also identifies areas where more research is needed. Building on this, the review sets up the following objectives and research questions.

Research Objectives

1) To identify the physiological signals that have been used in included studies of student mental-health screening.
2) To summarize machine learning models and mental health outcomes reported in the included studies.
3) To combine reported performance metrics and methodological limitations, and identify gaps for future research

Research Questions

1) Which physiological signals have been measured in studies and how have researchers applied them?
2) Which machine learning models and mental health outcomes are most frequently reported in the included studies?
3) Which performance metrics and methodological limitations do included studies report and what gaps remain for future research?

To answer these questions, a systematic review was conducted in accordance with the PRISMA 2020 guidelines. The following section describes the review period, criteria used for choosing studies and data extracted from the studies

II. Methodology

This systematic review was conducted in accordance with the PRISMA 2020 guidelines to ensure transparency and reproducibility. The review took place between September 30th and October 2nd, 2026 and was conducted by a single reviewer. Clear eligibility criteria were defined prior to the search, and the process involved systematically searching relevant databases, removing duplicates, and extracting structured

data from the included studies. A PRISMA flow diagram was used to document the number of records identified, screened, excluded, and included at each stage.

The literature search was carried out in PubMed and IEEE Xplore on 1 October 2026. These were chosen for their comprehensive coverage on biomedical, health, computing, and machine learning research. A Boolean search string was applied to capture studies that combined physiological or behavioral signals with machine learning approaches for mental health outcomes. The query used was: ("physiological signals" OR "behavioral data") AND ("Machine Learning" OR "Classification") AND ("depression" OR "anxiety disorder" OR "anxiety" OR "stress") AND ("Learning Institutions"). This strategy ensured that only studies relevant to student populations and academic contexts were retrieved. Studies were included if they:

- Reported quantitative results on physiological signals or behavioral data.
- Applied machine learning models, specifically Random Forest or comparable algorithms.
- Focused on mental health outcomes such as depression, anxiety, or stress in student or academic populations.
- Were peer-reviewed and published in English between 2018 and 2026. Earlier studies that were peer reviewed and had foundational information relevant to the review questions were also included.

Exclusion criteria were:

- Studies without empirical data or quantitative results.
- Opinion pieces, editorials, or reviews lacking methodological detail.
- Studies were excluded if they did not focus on the target population, mental-health outcomes, physiological/behavioral data, or machine-learning classification
- Records with abstracts in English and body in another language.

All records retrieved from PubMed and IEEE Xplore were first screened by title and abstract to assess relevance to the research objectives. Studies that met the inclusion criteria were retained for full-text review. Removal of duplicates were done by software suchas Mendeley based on doi. Text that passed the screening criteria underwent full text review and the were manually chosen for inclusion . . The PRISMA flow diagram summarizes the screening process: a total of 45 records were identified (31 from PubMed, 14 from IEEE Xplore). After removing duplicates (n=2), 43 records were screened. Twenty were excluded at the abstract stage, leaving 23 for fulltext retrieval. Three could not be retrieved, and of the 20 assessed for eligibility, three were excluded for relying solely on wearable sensors. Seventeen studies were ultimately included in the synthesis.

For each of the 17 included studies, relevant information was systematically extracted into a structured table. The extraction focused on variables that directly addressed the research objectives. The following data items were recorded:

- **Study details:** Author(s), year of publication, country of study.
- **Physiological signals:** Type of signals measured (e.g., heart rate, pupil size).
- **Algorithms applied:** Machine learning models used (Random Forest, Support Vector Machine(SVM),Convolutional Neural Networks(CNN), Decision Tree).
- **Outcomes:** Mental health states assessed (stress, anxiety, depression,Loneliness).
- **Performance metrics:** Accuracy, F1 score,precision and AUC-ROC.
- **Notes:** Limitations, strengths, or unique methodological contributions.

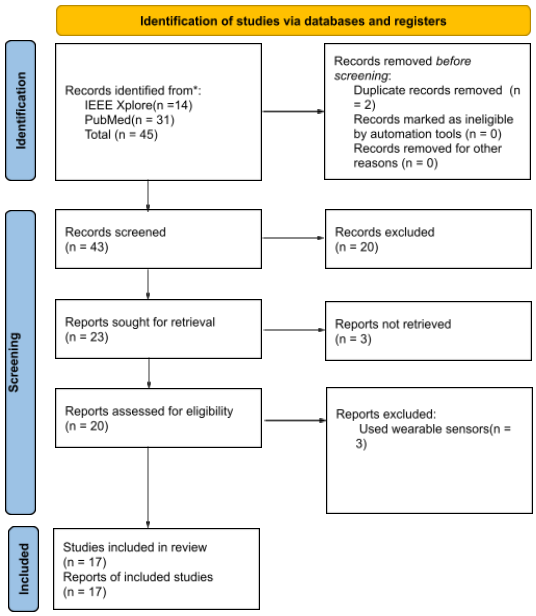


Fig. 1. PRISMA 2020 flow diagram of the study selection process.

Table 1: Summary of included studies (n=17)

Title / Focus	Year
A Multimodal Data Fusion Approach for Early Mental Health Risk Detection in College Students Using Logistic Regression and Random Forest	2026
The Predictive Model of Mental Illness using Decision Tree and Random Forest classification in Machine Learning	2022
Interpretable Hybrid Machine Learning for Student Mental Health Screening	2026
Algorithm Driven Early Warning and Intervention System for Mental Health of college students	2024
KD-EWS: An Explainable AI Framework for Early Warning and Decision Support in Academic Environments	2026
A psychologically interpretable artificial intelligence framework for the screening of loneliness, depression, and anxiety	2025
Use of physiological signals, behavioral data, and processing algorithms in electronic devices and mobile applications for diagnosing depression, anxiety, and stress	2026
Comparative Assessment of Machine Learning Models for Detection of Mental Health of Students: A Case Study of Undergraduate Students	2025
Human emotion recognition using Machine learning techniques based on the physiological signal	2025
A multimodal fairness aware machine learning framework for mental health risk prediction in university students	2026
Automated detection of mental disorders using physiological signals and machine learning: A systematic review and scientometric analysis	2023
Pupillometric measures of cognitive and emotional processes	2004
Correlates of behavioral and emotional disorders among school-going adolescents in Uganda	2025
Integrating Pupilometry and Self-Assessment for Holistic Evaluation of Metaverse Experiences	2023
Measuring task engagement as an input to physiological computing	2009
Pupil size as a biomarker of cognitive (dys)functions: Toward a physiologically informed screening of mental states	2025
Infrared Camera-Based Non-contact Measurement of Brain Activity From Pupillary Rhythms	2018

A. Risk of bias assessment.

Risk of bias was assessed by one reviewer using criteria covering data preparation, modelling and performance reporting. Particular attention was given to external validation. No assessment tool was used. The assessment informed the interpretation of findings but was not used to exclude studies.

III. Results

A. Characteristics of included studies.

The 17 included studies varied in their primary focus. To provide transparency, they were as follows:

- **Algorithmic performance:** 8 studies evaluated machine-learning models (e.g., Random Forest, Logistic Regression, Decision Trees, CNNs) for detecting mental-health outcomes.
- **Pupilometry:** 5 studies investigated pupil diameter as a physiological marker of cognitive load, stress, or emotional states.
- **General mental health context:** 2 studies examined general mental health among students within Uganda . No machine learning was used.
- **Explainable AI frameworks:** 2 studies explored explainable in academic environments.

B. Population size

Reported sample sizes varied across the studies. One study collected EEG and pupil response data from 70 volunteers [3]. The EmoLDANet study analyzed video recorded data from 50 participants to detect negative emotions from responses from their questionnaires [6]. A questionnaire based study included 142 students from a rural engineering institution covering aspects like sleep quality [7] and finally another survey used questionnaires including 1953 students aged 10-18 from schools in Central and eastern Uganda [2].

C. Physiological Signals

Physiological studies were reported unevenly across the studies. Pupil diameter mainly appeared in studies focused on pupilometry [3]–[5]. Generally the studies didnot report as many signals used for assessment. EEG appeared in a study that used the ASCERTAIN dataset [8]. The limited evidence limited comparison of signals across studies.

D. Mental Health Outcomes

The three main outcomes assessed were **stress**,**anxiety** and **depression** and appeared together in 6 studies [9], [10].

E. Algorithms

Across the included studies,a range of machine learning algorithms were evaluated for mental health screening. Logistic Regression, Decision Trees and Random Forests were among the traditional classifiers reported [9]–[11]. Logistic regression failed on complex datasets with class imbalances [9]. Decision trees in one particular study used for training and making predictions for target classes [11]. .

Beyond the standard traditional approaches, other studies incorporated advanced AI models. For example, architectures like the EmoLDANet were employed to sense negative thoughts through voice recognition [6]. The architecture used VGGII and Decision Trees and a knowledge driven .

F. Performance Metrics

A range of evaluation metrics such as accuracy, F1 score and Area Under the Curve(AUC-ROC) Were reported across the studies. All eight studies that used algorithms reported accuracy as their evaluation metric [6]–[8], [12], [13]. For example, an accuracy of 98.4% was reported for Logistic Regression [9], while another study reported 96.6% for Logistic Regression and 82.8% for Random Forest [10]. These values illustrate how accuracy was the primary metric used while evaluating traditional algorithms though using accuracy alone can be misleading in presence of class imbalance . Two studies used F1 score and Area Under the Curve(AUC) for model assessment [6], [14]. A study reported that an explainable AI achieved an F1 score of 80% while detecting loneliness, depression and anxiety [6]. Another study implemented a knowledge driven early warning system that achieved an F1 score 0.89 and AUC-ROC of 0.92 [14]. Because the studies used different outcomes, signals and metrics, these values do not establish that an algorithm out performed another. This limits comparison across studies.

IV. Discussion

Combination of evidence from different studies identified Random Forests as a good method for classification of signals showing their robustness in handling non-linear data. Relying on accuracy as the primary metric limits comparison across studies especially among those with imbalanced datasets. Stress, anxiety and depression appeared as the main mental health outcomes across most studies. Loneliness appeared in only one [6]. Traditional models such as logistic regression and decision trees are easily interpretable but are not scalable while deep learning models such as EmoLDANet and VGGII were used to identify emotion from speech recognition but are computationally extensive and may not be feasible in resource limited places.

Future research should emphasize multi modal approaches, standardized evaluation metrics and explainable AI to ensure reliable and practical deployment of physiological signal-based mental health screening

A. Limitations

This review identified several limitations across the included studies:

1) **Diversity of datasets and methods:** The studies used different performance metrics, outcomes, algorithms and datasets. This inhibited direct comparison of model performance
2) **Data acquisition challenges:** A study on pupilometry reported that while acquiring pupil movements [5]. Problems like light intensity, movement while recordings all affected readings.
3) **Metric reporting bias:** Majority of the studies used accuracy as their main performance metric. Sensitivity and precision were rarely included. Few studies used F1 score and AUC-ROC. Reliance on accuracy alone makes the model miss false positives and negatives, especially in imbalanced datasets.
4) **Sample size and representativeness:** Several studies used relatively small sample sizes particularly those carried out in controlled settings like laboratories. These samples may not fully represent the general population. Results from controlled settings may not fully transfer to educational settings such as schools.

V. Conclusion

This review examined how both physiological signals and machine learning methods have been used in the screening of mental health outcomes in educational institutions across 17 studies. Random Forest, Decision Trees and Logistic Regression were the most dominant traditional algorithms reported. Stress, anxiety and depression were primary outcomes across the studies. Accuracy was the most reported performance metric though two studies included AUC-ROC and F1 score limiting comparison across studies.

Population characteristics revealed a clear methodological divide. In some studies questionnaires were used and had large population sizes though they are prone to bias. By contrast, studies reported in controlled environments using EEG,pupilometry or facial recognition involved smaller populations reflecting the resource intensive nature of physiological data collection.

Overall, the included literature shows that machine learning approaches using physiological signals have been investigated for student mental health screening, but the findings do not yet support direct comparison across studies.

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